

Notes: Hirano, Imbens & Ridder (Econometrica, 2003)

Efficient Estimation of Average Treatment Effects Using the Estimated Propensity Score

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1 Big picture

- **Question.** How should researchers efficiently estimate average treatment effects when treatment assignment is unconfounded conditional on covariates?
- **Classic problem.** Inverse propensity-score weighting removes selection bias under unconfoundedness, but weighting by the *true* propensity score is generally not semiparametrically efficient.
- **Core result.** Weighting by the inverse of a *nonparametric estimate* of the propensity score, rather than by the true propensity score, achieves the semiparametric efficiency bound for several treatment-effect parameters.
- **Why surprising.** Estimation error in the propensity score is not merely a nuisance. When the score is estimated flexibly, the estimated weights impose covariate-balance information and remove inefficiency that remains under true weights.
- **Main estimands.** The paper studies the population average treatment effect (ATE), weighted average treatment effects (WATE), and the average treatment effect for the treated (ATT).
- **Technical route.** The propensity score is estimated by a series logit estimator, allowing the dimension of the approximation to grow with sample size.
- **Intuition.** The estimated-score weighting estimator can be interpreted as an empirical-likelihood or GMM estimator that incorporates the moment restrictions implied by treatment assignment probabilities.
- **Advantage over earlier efficient estimators.** The estimator requires nonparametric estimation of the propensity score only, while other efficient estimators require nonparametric estimation of outcome regression functions as well.
- **Empirical implication.** In empirical work, simply replacing the estimated propensity score with a known or true propensity score can reduce efficiency. Estimated scores can be statistically better because they enforce balancing restrictions in the sample.
- **Economic takeaway.** The propensity score is not just a dimension-reduction device for bias removal. It is also a vehicle for efficient estimation when estimated in a sufficiently flexible way.

2 Data, objects, and measurement

2.1 No empirical data table: the object is a semiparametric model

The paper is theoretical and econometric rather than empirical. It does not estimate treatment effects in a particular application. Instead, it studies a random sample of units with observed treatment status, outcome, and covariates, and derives large-sample efficiency properties for estimators.

Interpretation: The “data” in this paper are abstract random variables. The relevant measurement problem is how to transform observed (Y_i, T_i, X_i) into efficient estimates of causal parameters.

2.2 Potential outcomes and observed outcomes

For unit i , let:

- $T_i \in \{0, 1\}$ denote treatment status;
- $Y_i(1)$ denote the potential outcome under treatment;
- $Y_i(0)$ denote the potential outcome under control;
- X_i denote observed covariates.

The observed outcome is

$$Y_i = T_i Y_i(1) + (1 - T_i) Y_i(0).$$

Interpretation: The causal problem is missing data: each unit reveals only one potential outcome. The paper’s estimators use treatment assignment probabilities to reweight observed outcomes and reconstruct population means.

2.3 Estimands

The population average treatment effect is

$$\tau = E[Y(1) - Y(0)].$$

The weighted average treatment effect is

$$\tau_{\text{wate}} = \frac{\int E[Y(1) - Y(0) \mid X = x] g(x) dF(x)}{\int g(x) dF(x)},$$

where $g(x)$ is a known weight function.

The average treatment effect for the treated is

$$\tau_{\text{treated}} = E[Y(1) - Y(0) \mid T = 1].$$

Interpretation: The WATE nests many policy-relevant estimands. Setting $g(x) = p(x)$, where $p(x)$ is the propensity score, gives the ATT.

2.4 Unconfoundedness and overlap

The identifying assumption is

$$T \perp (Y(0), Y(1)) \mid X.$$

The propensity score is

$$p(x) = \Pr(T = 1 \mid X = x) = \mathbb{E}[T \mid X = x].$$

Rosenbaum and Rubin's balancing result implies

$$T \perp (Y(0), Y(1)) \mid p(X).$$

The paper also assumes overlap:

$$0 < \underline{p} \leq p^*(x) \leq \bar{p} < 1.$$

Interpretation: Unconfoundedness identifies causal effects; overlap ensures the inverse weights do not explode. The efficiency analysis asks which identified estimator extracts the most information from the same observed data.

2.5 Series logit propensity-score estimator

Let $R^K(x) = (r_{1K}(x), \dots, r_{KK}(x))'$ be a growing vector of basis functions. The series logit estimator is

$$\hat{p}(x) = L(R^K(x)' \hat{\pi}_K), \quad L(a) = \frac{\exp(a)}{1 + \exp(a)},$$

where

$$\hat{\pi}_K = \arg \max_{\pi} \sum_{i=1}^N [T_i \log L(R^K(X_i)' \pi) + (1 - T_i) \log \{1 - L(R^K(X_i)' \pi)\}].$$

Interpretation: The logistic form respects the fact that a propensity score is a probability. The growing series makes the estimator nonparametric in large samples.

3 Models and empirical specifications

3.1 Baseline inverse-propensity-weighted ATE estimator

Given an estimator $\hat{p}(X_i)$, the ATE estimator is

$$\hat{\tau} = \frac{1}{N} \sum_{i=1}^N \left[\frac{Y_i T_i}{\hat{p}(X_i)} - \frac{Y_i (1 - T_i)}{1 - \hat{p}(X_i)} \right].$$

Interpretation: The treated observations are weighted by the inverse probability of receiving treatment; controls are weighted by the inverse probability of receiving control. This reconstructs the two potential-outcome means under unconfoundedness.

3.2 Simple missing-data example: true weights versus estimated weights

The paper first gives intuition in a missing-data problem with a binary covariate and known selection probability $p(x) = 1/2$. The variance bound is

$$V_{\text{bound}} = 2 \mathbb{E}[\text{Var}(Y \mid X)] + \text{Var}(\mathbb{E}[Y \mid X]).$$

Using true weights gives

$$V_{\text{tw}} = V_{\text{bound}} + \mathbb{E}[\mathbb{E}(Y \mid X)^2],$$

which is larger unless $\mathbb{E}[Y \mid X] = 0$. Using estimated cell probabilities gives

$$V_{\text{ew}} = V_{\text{bound}}.$$

Interpretation: Estimating the weights forces the weighted sample to obey the observed covariate distribution. True weights ignore this extra sample information, so they leave variance on the table.

3.3 GMM and empirical-likelihood intuition

The true-weight estimator uses the moment condition

$$\mathbb{E} \left[\frac{YT}{p(X)} - \theta \right] = 0.$$

But the known or estimated propensity score also implies balance moments such as

$$\mathbb{E}[(T - p(X))h(X)] = 0$$

for functions $h(X)$.

Interpretation: Estimated weights implicitly use these balance restrictions. This is why the estimator behaves like an empirical-likelihood estimator: it chooses weights that both estimate the causal parameter and respect treatment-assignment information.

3.4 Asymptotic linear representation for the ATE

The paper shows that the estimated-score estimator is asymptotically linear:

$$\hat{\tau} = \tau^* + \frac{1}{N} \sum_{i=1}^N \{\psi(Y_i, T_i, X_i, \tau^*, p^*(X_i)) + \alpha(T_i, X_i)\} + o_p(N^{-1/2}),$$

where the adjustment term is

$$\alpha(t, x) = - \left(\frac{\mu_1(x)}{p^*(x)} + \frac{\mu_0(x)}{1 - p^*(x)} \right) (t - p^*(x)).$$

Interpretation: The adjustment term is the efficiency gain from estimating the propensity score. It subtracts the component of the score that can be predicted from the treatment-assignment process.

3.5 Theorem 1: efficient ATE estimation

Under Assumptions 1–5,

$$\sqrt{N}(\hat{\tau} - \tau^*) \Rightarrow \mathcal{N}(0, V),$$

with

$$V = \mathbb{E} \left[(\tau(X) - \tau)^2 + \frac{\sigma_1^2(X)}{p^*(X)} + \frac{\sigma_0^2(X)}{1 - p^*(X)} \right].$$

The estimator reaches the semiparametric efficiency bound.

Interpretation: This is the central formal result. Estimating the propensity score nonparametrically makes the inverse-weighting estimator as efficient as possible in the semiparametric model.

3.6 Weighted average treatment effects

For a known weight function $g(x)$, the paper studies

$$\tau_{\text{wate}} = \frac{\text{E}[\tau(X)g(X)]}{\text{E}[g(X)]}.$$

The corresponding efficient variance bound is

$$V = \frac{1}{\text{E}[g(X)]^2} \text{E} \left[g(X)^2 (\tau(X) - \tau_{\text{wate}})^2 + \frac{g(X)^2}{p^*(X)} \sigma_1^2(X) + \frac{g(X)^2}{1 - p^*(X)} \sigma_0^2(X) \right].$$

Interpretation: The same logic extends to subpopulation-weighted effects. Choosing g can also reduce overlap problems by focusing on covariate regions with sufficient treated-control overlap.

3.7 Average treatment effect for the treated

The ATT is a special WATE with $g(x) = p^*(x)$:

$$\tau_{\text{treated}} = \frac{\text{E}[\tau(X)p^*(X)]}{\text{E}[p^*(X)]}.$$

If the propensity score is known, the efficient estimator uses $p^*(X)$ as the external weight but uses $\hat{p}(X)$ in the inverse-probability denominators.

If the propensity score is unknown, replacing $p^*(X)$ everywhere with $\hat{p}(X)$ yields an efficient estimator for ATT.

Interpretation: ATT is more subtle than ATE because knowledge of the propensity score affects the efficiency bound. The paper clarifies when true and estimated scores should appear in different parts of the estimator.

4 Identification and theoretical design

4.1 What identifies the causal effect

- Unconfoundedness identifies treatment effects from observed conditional mean differences.
- Propensity-score balancing allows adjustment using $p(X)$ rather than the full covariate vector.
- Overlap makes the weighted averages well behaved.

Interpretation: Identification is not the novel contribution. The novelty is efficiency: once the parameter is identified, which weighting estimator uses the available information best?

4.2 Why estimated propensity scores help

- Estimating $p(X)$ uses the realized joint distribution of treatment and covariates.
- The resulting weights impose balance restrictions between treatment status and covariates.
- These restrictions remove noise that remains in the true-weight estimator.

Interpretation: In finite samples, researchers often think of propensity-score estimation as an approximation burden. The paper shows that, asymptotically, the estimation step can be an efficiency device.

4.3 Why nonparametric estimation matters

Parametric propensity-score models can improve efficiency if correctly specified, but misspecification can introduce bias or fail to achieve the efficiency bound. A growing series logit estimator avoids treating a fixed finite-dimensional logit as the true model.

Interpretation: The efficiency result is a semiparametric result. The flexibility of the first-stage estimator is essential.

4.4 What the paper cannot fully answer

- The paper's main results are asymptotic; finite-sample performance is explicitly left open.
- Sieve estimation can be difficult in high dimensions.
- The theory assumes unconfoundedness, overlap, and smoothness.
- The result does not make inverse weighting robust to misspecified covariates or unobserved selection.
- Modern machine-learning first stages require separate regularity conditions not developed in the original paper.

Interpretation: The paper is a foundational efficiency result, not a complete practical recipe for every empirical application.

5 Equation and theorem exhibits

5.1 Exhibit 1: estimands and unconfoundedness

Read:

- The exhibit defines the ATE, WATE, and ATT.
- It also states the unconfoundedness assumption.
- The ATT appears as a special WATE with $g(x) = p(x)$.

Interpretation: The exhibit separates two problems. First, treatment effects must be identified using unconfoundedness. Second, once identified, the paper asks how to estimate them efficiently. The ATE, WATE, and ATT differ only in how covariate-specific treatment effects are weighted.

Economic message: The choice of estimand changes the population being summarized. In empirical work, this distinction matters as much as the estimator: ATE, WATE, and ATT can answer different policy questions.

$T_i \cdot Y_i(1) + (1 - T_i) \cdot Y_i(0)$. In addition, we observe a vector of covariates denoted by X_i .⁴ Initially we focus on the population average treatment effect:

$$(1) \quad \tau \equiv \mathbf{E}[Y(1) - Y(0)].$$

We shall also discuss estimation of weighted average treatment effects,

$$(2) \quad \tau_{\text{wate}} \equiv \frac{\int \mathbf{E}[Y(1) - Y(0)|X = x]g(x)dF(x)}{\int g(x)dF(x)},$$

where $g(\cdot)$ is a known function of the covariates.⁵ In the special case where the weight function $g(x)$ is equal to the propensity score $p(x) = \Pr(T = 1|X = x)$, this leads under the unconfoundedness assumption to the average effect for the treated:

$$(3) \quad \tau_{\text{treated}} \equiv \mathbf{E}[Y(1) - Y(0)|T = 1].$$

The central problem of evaluation research is that for unit i we observe either $Y_i(0)$ or $Y_i(1)$, but never both. To solve the identification problem, we maintain throughout the paper the unconfoundedness assumption (Rubin (1978), Rosenbaum and Rubin (1983)), related to the selection-on-observables assumption (Barnow, Cain, and Goldberger (1980)), which asserts that conditional on the observed covariates, the treatment indicator is independent of the potential outcomes. Formally, we have the following assumption:

ASSUMPTION 1 (Unconfounded Treatment Assignment):

$$T \perp (Y(0), Y(1)) \mid X.$$

Heckman, Ichimura, and Todd (1998) point out that for identification of the average treatment effect τ this assumption can be weakened to mean independence ($\mathbf{E}[Y(t)|T, X] = \mathbf{E}[Y(t)|X]$ for $t = 0, 1$). If one is interested in the average effect for the treated, the assumption can be further weakened to only require

³ Implicit in this notation is the stability assumption or SUTVA (Rubin (1978)) that units are not affected by receipt of treatment by others, and that there is only one version of the treatment.

⁴ These variables are assumed not to be affected by the treatment.

⁵ An alternative estimand, which we do not consider here, is the direct weighted average treatment effect of the form

$$\tau_{\text{dwate}} = \frac{\int \mathbf{E}[Y(1) - Y(0)|X = x]g(x) dx}{\int g(x) dx},$$

Original-paper exhibit: estimands and unconfoundedness.

5.2 Exhibits 2 and 3: why estimated weights beat true weights

Read:

- Exhibit 2 presents the true-weight estimator and its variance in the simple missing-data example.

- The true-weight estimator has an extra variance term, $E[E(Y | X)^2]$.
- Exhibit 3 shows the estimated-weight estimator reaches the variance bound.
- The GMM interpretation shows that treatment-assignment information supplies additional moment restrictions.

Interpretation: The exhibits explain the paper’s most counterintuitive point. The true selection probability is not automatically the best object to plug into inverse weights. Estimated weights use the empirical treatment distribution within covariate cells, which absorbs predictable variation and enforces balance.

Economic message: The statistical lesson is analogous to using controls in a randomized experiment: even when randomization probabilities are known, conditioning on realized covariate balance can improve precision.

observations with $T_i = t$ and $X_i = x$, for $t, x \in \{0, 1\}$. Furthermore, suppose the true selection probability is constant, $p(x) = 1/2$ for all $x \in \{0, 1\}$.⁸ The normalized variance bound for β_0 is

$$(6) \quad V_{bound} = 2 \cdot \mathbf{E}[\mathbf{V}(Y|X)] + \mathbf{V}(\mathbf{E}[Y|X]).$$

The “true weights” estimator weights the complete observations by the inverse of the true selection probability:

$$(7) \quad \hat{\beta}_{tw} = \frac{1}{N} \sum_{i=1}^N \frac{Y_i \cdot T_i}{p(X_i)} = \frac{1}{N} \sum_{i=1}^N \frac{Y_i \cdot T_i}{1/2}.$$

Its large sample normalized variance is

$$V_{tw} = 2 \cdot \mathbf{E}[\mathbf{V}(Y|X)] + \mathbf{V}(\mathbf{E}[Y|X]) + \mathbf{E}[\mathbf{E}[Y|X]^2] = V_{bound} + \mathbf{E}[\mathbf{E}[Y|X]^2],$$

strictly larger than the variance bound (6) unless $\mathbf{E}[Y|X] = 0$.

The second estimator weights the complete observations by the inverse of a nonparametric estimate of the selection probability. This estimator is the main focus of the paper and it will be discussed in Section 4 in more general settings. In the current setting the estimated selection probability is simply the proportion of observed outcomes for a given value of the covariate. For units with $X_i = 0$ the proportion of observed outcomes is $N_{10}/(N_{00} + N_{10})$, and for units with $X_i = 1$ the proportion of observed outcomes is $N_{11}/(N_{01} + N_{11})$. Thus the estimated selection probability is

$$\hat{p}(x) = \begin{cases} N_{10}/(N_{00} + N_{10}) & \text{if } x = 0, \\ N_{11}/(N_{01} + N_{11}) & \text{if } x = 1. \end{cases}$$

The proposed “estimated weights” estimator is then

$$(8) \quad \hat{\beta}_{ew} = \frac{1}{N} \sum_{i=1}^N \frac{Y_i \cdot T_i}{\hat{p}(X_i)}.$$

⁷ An efficient estimator is easily obtained by averaging the within-subsample difference of treatment/control averages. It can also be found by specializing the more general estimators in Robins and Rotnitzky (1995) and Hahn (1998) to this simple case. The discussion here is solely intended to convey intuition for the formal results that will be presented in Section 4.

⁸ Thus the missing data are Missing Completely At Random (MCAR; Rubin (1976), Little and Rubin (1987)).

The normalized variance of this estimator is equal to the variance bound:

$$V_{ew} = 2 \cdot \mathbf{E}[\mathbf{V}(Y|X)] + \mathbf{V}(\mathbf{E}[Y|X]) = V_{bound}.$$

Not only does the weighting estimator with nonparametrically estimated weights have a lower variance than the estimator using the “true” weights in this simple case, but it is in fact fully efficient. In the remainder of this section we shall provide some intuition for this result. This will suggest why this efficiency property may carry over to the case with continuous and vector-valued covariates, as well as with general dependence of the selection probability or propensity score on the covariates.

An alternative interpretation of the estimated-weights estimator is based on a Generalized Method of Moments (GMM) representation (Hansen (1982)). Under the assumption that the selection probability is $p(x) = 1/2$, we can estimate β_0 using the single moment restriction $\mathbf{E}[\psi_1(Y, X, T, \beta_0)] = 0$, with

$$\psi_1(y, t, x, \beta) = \frac{y \cdot t}{p(x)} - \beta = \frac{y \cdot t}{1/2} - \beta.$$

The GMM estimator based on the single moment restriction $\psi_1(\cdot)$, given knowledge of the selection probability, is the true-weights estimator $\hat{\beta}_{tw}$ in (7). However, this estimator is not necessarily efficient, because it ignores the additional information that is available in the form of knowledge of the selection probability. This additional information can be written in moment condition form as $\mathbf{E}[T - p(X)|X] = \mathbf{E}[T - 1/2|X] = 0$. With a binary covariate this conditional moment restriction corresponds to two marginal moment restrictions, $\mathbf{E}[\psi_2(Y, T, X, \beta_0)] = 0$, with:

$$\psi_2(y, t, x, \beta) = \begin{pmatrix} x \cdot (t - 1/2) \\ (1 - x) \cdot (t - 1/2) \end{pmatrix}.$$

Estimating β_0 in a generalized method of moments framework using the moments $\psi_1(\cdot)$ and $\psi_2(\cdot)$ leads to a fully efficient estimator.⁹ Here it is of partic-

- (a) True weights and estimated weights in the simple example.

(b) Estimated weights, GMM moments, and efficiency intuition.

Original-paper exhibits: the estimated-weight intuition.

5.3 Exhibit 4: series logit propensity-score estimation

Read:

- The exhibit defines the series logit estimator for $p^*(x)$.
- The logistic function keeps the estimated propensity score between zero and one.
- The number of terms grows with sample size, making the estimator nonparametric.

Interpretation: A linear series approximation could predict probabilities outside $[0, 1]$. The series logit construction preserves probability structure while retaining flexibility.

Economic message: The first stage is not just a practical nuisance. It is the mechanism through which covariate-balance information enters the efficient weighting estimator.

and satisfies $|\lambda(k)| \leq |\lambda(k+1)|$, and let $x^\lambda = \prod_{j=1}^r x_j^{\lambda_j}$. For a sequence $\lambda(k)$ we consider the series $r_{kK}(x) = x^{\lambda(k)}$. If we denote the logistic cdf by $L(a) = \exp(a)/(1 + \exp(a))$, the SLE for $p^*(x)$ is defined by $\hat{p}(x) = L(R^K(x)' \hat{\pi}_K)$ with

$$\hat{\pi}_K = \arg \max_{\pi} \sum_{i=1}^N (T_i \cdot \ln L(R^K(X_i)' \pi) + (1 - T_i) \cdot \ln(1 - L(R^K(X_i)' \pi))).$$

In Appendix A we discuss the relevant asymptotic theory for $\hat{p}(x)$.

In addition to the unconfoundedness assumption the following assumptions are used to derive the properties of the estimator. First, we restrict the distribution of X , $Y(0)$, and $Y(1)$:

ASSUMPTION 2 (Distribution of X):
(i) the support $\mathbf{X}$ of the r -dimensional covariate X is a Cartesian product of compact intervals, $\mathbf{X} = \prod_{j=1}^r [x_{lj}, x_{uj}]$;
(ii) the density of X is bounded, and bounded away from 0, on $\mathbf{X}$.

ASSUMPTION 3 (Distribution of $Y(0)$, $Y(1)$):
(i) $\mathbf{E}[Y(0)^2] < \infty$ and $\mathbf{E}[Y(1)^2] < \infty$;
(ii) $\mu_0(x)$ and $\mu_1(x)$ are continuously differentiable for all $x \in \mathbf{X}$.

The next assumption requires sufficient smoothness of the propensity score.

ASSUMPTION 4 (Selection Probability): The propensity score $p^*(x)$ satisfies the

Original-paper exhibit: series logit propensity-score estimator.

5.4 Exhibit 5: Theorem 1 for efficient ATE estimation

Read:

- Theorem 1 states consistency, asymptotic normality, and efficiency for the ATE estimator using the estimated propensity score.
- The variance decomposes into treatment-effect heterogeneity and the two conditional outcome variances weighted by inverse treatment probabilities.
- The exhibit also shows the adjustment term from estimating $p^*(X)$.

Interpretation: The adjustment term is the formal expression for the efficiency gain. It removes from the influence function the part that can be predicted using treatment-assignment information.

Economic message: In empirical terms, the theorem says that a carefully estimated propensity score can deliver the same asymptotic precision as the best possible semiparametric estimator for the ATE.

THEOREM 1: Suppose Assumptions 1–5 hold. Then:

- (i) $\hat{\tau} \xrightarrow{p} \tau^*$;
- (ii) $\sqrt{N}(\hat{\tau} - \tau^*) \xrightarrow{d} \mathcal{N}(0, V)$, where

$$\begin{aligned} V &= \mathbf{E} \left[\left(\left(\frac{YT}{p^*(X)} - \frac{Y(1-T)}{1-p^*(X)} - \tau^* \right) \right. \right. \\ &\quad \left. \left. - \left(\frac{\mu_1(X)}{p^*(X)} + \frac{\mu_0(X)}{1-p^*(X)} \right) (T - p^*(X)) \right)^2 \right] \\ &= \mathbf{E} \left[(\tau(X) - \tau)^2 + \frac{\sigma_1^2(X)}{p^*(X)} + \frac{\sigma_0^2(X)}{1-p^*(X)} \right]; \text{ and} \end{aligned}$$

- (iii) $\hat{\tau}$ reaches the semiparametric efficiency bound.

PROOF: See Appendix B.

REMARK 1: This result also covers the extension of the binary-covariate MAR example in Section 3 to the continuous covariate case. For this case set $Y = 0$ if $T = 0$ and set $Y(0)$ identically equal to 0.

REMARK 2: Theorem 1 establishes the result for continuous X . If X has both continuous and discrete components, this can be dealt with in a conceptually straightforward manner by using the continuous covariate estimator within sam-

Original-paper exhibit: Theorem 1 and the ATE efficiency result.

5.5 Exhibits 6 and 7: WATE and ATT

Read:

- Exhibit 6 gives the efficiency bound for WATE and shows that ATT is obtained by setting $g(x) = p^*(x)$.
- Exhibit 7 gives the efficient ATT result when the propensity score is known.
- The ATT estimator uses the true propensity score as the external population weight, but uses the estimated score in the inverse-weight denominators.

Interpretation: The ATT case is delicate because the propensity score appears in two different roles: as a definition of the target population and as a selection probability. The estimator treats these roles differently to achieve efficiency.

Economic message: This is important for applied work because ATT is common in policy evaluation. The theorem warns against mechanically using either true or estimated scores everywhere without considering which role the score plays.

$g(x)/p^*(x)$ and $g(x)/(1-p^*(x))$ are bounded from above. Thus, if there is insufficient overlap in the distributions of the treated and untreated units, one may wish to choose $g(\cdot)$ to restrict attention to a subpopulation for which there is sufficiently large probability of observing both treated and untreated units.

A semiparametric efficiency bound for τ_{wate} has not been previously calculated in the literature. The next result shows that our estimator is efficient.

THEOREM 4: *The semiparametric efficiency bound for estimation of τ_{wate} is*

$$V = \frac{1}{\mathbf{E}[g(X)]^2} \mathbf{E} \left[g(X)^2 (\tau(X) - \tau_{\text{wate}})^2 + \frac{g(X)^2}{p^*(X)} \sigma_1^2(X) + \frac{g(X)^2}{1-p^*(X)} \sigma_0^2(X) \right].$$

PROOF: See Appendix B.

4.3. Estimating the Average Treatment Effect for the Treated

Under unconfoundedness the average treatment effect for the treated (Rubin (1977), Heckman and Robb (1985), Heckman, Ichimura, and Todd (1997, 1998)) is a special case of the weighted average treatment effect, corresponding to the weighting function $g(x) = p^*(x)$. To see this, first note that under unconfoundedness

$$\begin{aligned} \tau_{\text{treated}}^* &= \mathbf{E}[Y(1) - Y(0)|T = 1] = \mathbf{E}[\mathbf{E}[Y(1) - Y(0)|X, T = 1]|T = 1] \\ &= \mathbf{E}[\mathbf{E}[Y(1) - Y(0)|X]|T = 1] = \mathbf{E}[\tau(X)|T = 1]. \end{aligned}$$

Second, the latter is equal to

$$\mathbf{E}[\tau(X)|T = 1] = \int \tau(x) dF(x|T = 1)$$

(a) WATE efficiency and ATT as a weighted effect.

The estimator is the solution to

$$(18) \quad 0 = \sum_{i=1}^N p^*(X_i) \cdot \left(\frac{Y_i \cdot T_i}{\hat{p}(X_i)} - \frac{Y_i \cdot (1 - T_i)}{1 - \hat{p}(X_i)} - \tau_{\text{treated}} \right),$$

with the same nonparametric series estimator $\hat{p}(x)$ as before.

The next result, which follows directly from Theorem 4, shows that this estimator achieves the efficiency bound calculated by Hahn (1998) for estimation of the effect of treatment on the treated, assuming that the propensity score is known.

COROLLARY 1: *Suppose that Assumptions 1–5 hold. Then:*

- (i) $\hat{\tau}_{\text{treated}} \xrightarrow{p} \tau_{\text{treated}}^*$;
- (ii) $\sqrt{N}(\hat{\tau}_{\text{treated}} - \tau_{\text{treated}}^*) \xrightarrow{d} \mathcal{N}(0, V)$, with

$$V = \frac{1}{\mathbf{E}[p^*(X)]^2} \mathbf{E} \left[p^*(X)^2 (\tau(X) - \tau_{\text{treated}})^2 + p^*(X) \sigma_1^2(X) + \frac{p^*(X)^2}{1-p^*(X)} \sigma_0^2(X) \right], \text{ and}$$

- (iii) $\hat{\tau}_{\text{treated}}$ achieves the semiparametric efficiency bound.

The proof for this corollary is omitted as the result directly follows from Theorem 4.

Note that in the moment function (17) the propensity score appears in two places, first as $p^*(x)$ multiplying the remainder of the moment function where it replaces the general weight function $g(x)$ in (16), and second as $p(x)$ in the denominator of the two terms. We only use the estimated propensity score in the second part in the efficient estimator in (18). The result of the theorem above implies that this is more efficient than using the true propensity score everywhere and solving

$$(19) \quad 0 = \sum_{i=1}^N p^*(X_i) \cdot \left(\frac{Y_i \cdot T_i}{p^*(X_i)} - \frac{Y_i \cdot (1 - T_i)}{1 - p^*(X_i)} - \tau_{\text{treated}} \right),$$

or using the estimated propensity score everywhere, which amounts to solving

(b) ATT efficiency when the propensity score is known.

Original-paper exhibits: WATE and ATT results.

5.6 Exhibit 8: ATT when the propensity score is unknown

Read:

- Theorem 5 covers the case in which the propensity score is not known.
- Replacing the propensity score with its estimated counterpart yields an efficient ATT estimator under the stated assumptions.
- The conclusion summarizes the paper's main message: true-score weighting is not efficient, but flexible estimated-score weighting is.

Interpretation: The theorem completes the paper's logic. Whether the propensity score is known by design or unknown and estimated from data, the efficient use of the score requires understanding how estimation interacts with the target parameter.

Economic message: The result directly informs modern causal inference: first-stage propensity-score estimation can be part of efficient estimation, not merely a source of generated-regressor uncertainty.

If the propensity score is not known, then Hahn (1998) shows that this affects the efficiency bound for the effect of treatment on the treated. Our previous estimator $\hat{\tau}_{treated}$ cannot be used since it makes use of $p^*(x)$. However, we can use the estimated propensity score in place of $p^*(x)$ in the weighting of observations as in (20). Call this estimator $\hat{\tau}_{te}$. The next theorem shows that this estimator is efficient if the propensity is not known.

THEOREM 5: Suppose that Assumptions 1–5 hold. Then:

- (i) $\hat{\tau}_{te} \xrightarrow{p} \tau_{treated}^*$;
- (ii) $\sqrt{N}(\hat{\tau}_{te} - \tau_{treated}^*) \xrightarrow{d} \mathcal{N}(0, V)$, with

$$V = \frac{1}{\mathbb{E}[p^*(X)]^2} \mathbb{E} \left[p^*(X)(\tau(X) - \tau_{treated}^*)^2 + p^*(X)\sigma_1^2(X) + \frac{p^*(X)^2}{1 - p^*(X)}\sigma_0^2(X) \right], \text{ and}$$

- (iii) $\hat{\tau}_{te}$ achieves the semiparametric efficiency bound for estimation of $\tau_{treated}$ when the propensity score is not known.

The proof goes along the same lines as that for Theorems 1 and 2 and is omitted.

5. CONCLUSION

In this paper we have studied efficient estimation of various average treatment effects under an unconfounded treatment assignment assumption. Although weighting observations by the inverse of the true propensity score does not lead to efficient estimators, weighting each observation by the inverse of a nonparametric estimate of the propensity score does lead to efficient estimators. We provide intuition for this result through connections to the literatures on empirical likelihood estimators and choice-based sampling.

The estimators proposed in this paper require fewer functions to be estimated nonparametrically than other efficient estimators previously proposed in the literature. Which estimators have more attractive finite sample properties, and which have more attractive computational properties, remain open questions.

Original-paper exhibit: Theorem 5 and the conclusion.

6 Short summary

- The paper studies efficient estimation of causal treatment effects under unconfoundedness.
- The main parameters are ATE, WATE, and ATT.
- Propensity-score adjustment removes bias, but true-score inverse weighting is not generally efficient.
- The paper shows that inverse weighting with a flexible nonparametric estimate of the propensity score can achieve the semiparametric efficiency bound.
- A simple missing-data example explains why: estimated weights use covariate-balance information that true weights ignore.
- The formal estimator uses a series logit propensity-score estimator.
- The ATE result achieves the same efficiency bound derived in the semiparametric literature.
- The WATE result extends the logic to subpopulation-weighted estimands.
- The ATT result is more subtle because the propensity score both defines the target population and appears in inverse weights.
- The estimators require fewer nuisance functions than some previous efficient estimators, because only the propensity score is estimated nonparametrically.
- The paper leaves finite-sample and computational comparisons as open questions.
- The enduring lesson is that estimated propensity scores can be more efficient than true propensity scores in inverse-weighted causal estimation.

7 Conclusion

7.1 Core conclusion

Hirano, Imbens, and Ridder show that the estimated propensity score has a deeper role than bias adjustment. Under unconfoundedness and regularity conditions, inverse weighting by a nonparametrically estimated propensity score yields semiparametrically efficient estimators of average treatment effects. In contrast, inverse weighting by the true propensity score generally fails to achieve the efficiency bound.

7.2 Economic intuition

The key intuition is that estimating the propensity score forces the weights to respect information contained in the realized treatment-covariate distribution. True weights treat the assignment probabilities as external and do not use all sample information. Estimated weights, especially nonparametric estimated weights, behave as if they impose balancing moment restrictions. These restrictions reduce avoidable variance.

7.3 Evidence chain inside the paper

The paper builds its argument in four steps. First, it defines the causal estimands and invokes unconfoundedness for identification. Second, it uses a simple binary-covariate missing-data example to show why estimated weights can dominate true weights. Third, it formalizes the estimator with a series logit first stage and derives the ATE efficiency result. Fourth, it extends the logic to WATE and ATT, showing how the role of the propensity score changes with the estimand.

7.4 Practical implications

For applied researchers, the paper cautions against treating the true propensity score as automatically optimal. Even in randomized experiments with known assignment probabilities, using an estimated or calibrated propensity-score model can improve precision by exploiting covariate information. In observational studies, the result supports flexible first-stage estimation as part of efficient causal estimation, provided the assumptions and rate conditions are credible.

7.5 What not to overinterpret

The efficiency result does not solve the problem of omitted confounders. It does not guarantee good finite-sample behavior when overlap is weak or weights are unstable. It does not imply that any estimated propensity score is efficient; the estimator must be sufficiently flexible and regular. It also does not automatically cover modern high-dimensional machine-learning first stages without additional theory.

7.6 Final takeaway

The paper’s enduring message is that propensity scores are both balancing objects and efficiency devices. The act of estimating them can improve the estimator because it incorporates information about assignment and covariate balance. This insight explains why generated propensity scores can outperform true propensity scores and why the first stage is central to efficient treatment-effect estimation.